Real-Time Monitoring Dashboard for Red-Light Camera Violations in Ottawa

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***Abstract*—Red-Light violations are a traffic concern in the City of Ottawa. There are datasets on red-light camera violations and locations that are available to the public, but they are difficult to analyze and gain insights from as they are not consolidated. The project develops Real-Time Monitoring Dashboard for Red-Light Camera Violations in Ottawa. Datasets from 2015 to 2026 were cleaned and transformed using Excel and Power Query. Then analyzed and visualized in Tableau. The resulting four-page interactive dashboard enables users to identify high-risk intersections, analyze temporal and seasonal trends, and detect anomalies in violation patterns across 86 red-light cameras. This supports more informed traffic safety decisions for city officials.**

Keywords—Ottawa, Red-Light Violations, Data Analysis

I. Introduction

Red-light camera violations are a significant traffic concern, and the City of Ottawa has public datasets on these red-light violations. However, they are not easily analyzed as data is spread in multiple datasets, so city officials may find it difficult to identify high-risk intersections or detect trends to inform traffic safety decisions.

The project addresses the knowledge gap by developing a Real-Time Monitoring Dashboard for Red-Light Camera Violations in Ottawa. The goal is to use the publicly available red-light camera violations and locations datasets from Open Ottawa to develop an interactive dashboard that identifies the high-risk intersections, analyzes the trends, and detects the anomalies in violation patterns across the 86 current red-light cameras.

II. Data Source

The datasets chosen are from the City of Ottawa's Open Ottawa (open data) website, where various datasets are available to the public to support participation and civic engagement.

A. Red-Light Camera Violations

The Red-Light Camera Violations datasets span the years 2015 to 2026. They provide information on red-light camera intersection, coordinates, installation year, camera facing direction, monthly violations and summary statistics. Each dataset has 21 columns and has up to 86 rows.

B. Red-Light Camera Locations

The Red-Light Camera Locations dataset provides information on all the red-light cameras installed in the city, such as the intersection, coordinates, installation year and camera facing direction. The dataset has 8 columns and 88 rows.

C. Traffic Camera Locations

The City of Ottawa maintains a publicly accessible Traffic Camera Locations website that provides live image feeds from intersections across the city.

III. Data Collection

Data collection involved downloading the datasets from the City of Ottawa's open data website. A total of 13 datasets in Comma-Separated Values (CSV) format were collected. The Red-Light Camera Violations datasets contained multiple columns that were also in the Red-Light Camera Locations dataset, making them redundant. Additionally, the Red-Light Locations dataset included two rows of notes at the end of the dataset, which would not be needed for analysis. These issues were later addressed during the data preparation and cleaning process.

In cases where an intersection is equipped with both a traffic camera and a red-light camera, the corresponding traffic camera feed URL from the Traffic Camera Locations website was identified and mapped to the relevant entry in the Red-Light Camera Locations table. This attribute was added as an additional field to the camera reference table, enabling the dashboard to surface real-time visual context for any selected intersection directly within Tableau.

IV. Data Preparation and Cleaning

All data preparation and cleaning steps were performed in Microsoft Excel prior to loading into Tableau. The following steps were applied systematically across all thirteen source files.

The twelve annual violation files were first consolidated into a single Excel workbook, with each year retained as a separate worksheet.

The camera reference table was cleaned by removing the French location description column, as only the English intersection names were required for the dashboard. The projected coordinate columns (X and Y) were also removed, as geographic mapping relied on the decimal degree latitude and longitude values already present in the dataset.

To enable joining of camera identifiers across the violation files and the reference table, a temporary composite key was constructed in each sheet. This key was formed by concatenating the camera installation year, latitude rounded to five decimal places, longitude rounded to five decimal places, and the camera facing direction. Rounding to five decimal places was applied to ensure consistent matching across files where minor floating-point variations existed in the coordinate values. This approach was necessary because some intersections share the same physical location but have cameras facing different directions, for example, an intersection may have both an Eastbound and a Westbound camera, making a coordinate-only key insufficient for unique identification.

The projected coordinate columns (X and Y) were then removed from all twelve violation files, as they were redundant given the retained decimal degree coordinates.

Using the composite temporary key, a camera identifier was assigned to each row in all twelve violation files via XLOOKUP against the camera reference table. This established a consistent foreign key relationship between the violation records and the camera metadata.

Following the successful assignment of camera identifiers, the temporary composite key column was removed from all sheets to eliminate redundancy.

A year column was added to each violation worksheet, populated with the corresponding calendar year of that file. This was required because the annual files did not contain an explicit year field, and the year value was needed to construct the final unified date dimension after unpivoting.

Remaining redundant columns were reviewed and removed across all sheets, retaining only the fields required for analysis: camera identifier, intersection name, installation year, latitude, longitude, facing direction, and the twelve monthly violation counts.

Finally, each of the twelve annual violation worksheets was unpivoted using Power Query in Excel. The wide-format monthly columns (January through December) were transformed into a long-format structure, producing a unified violations table with one row per camera per month. The resulting table contains four core fields: camera identifier, year, month, and total violations for that month. A new unique primary key was generated for each row to support record-level identification in Tableau. This final structure formed the basis of all calculated fields, KPI metrics, and visualizations built in the dashboard.

V. DATA MODELING

Tableau was chosen to visualize the data as it provides easy drag-and-drop analysis, no need for complex code, and can create stories.

Once the cleaned datasets were loaded into Tableau, a union was performed on the yearly file to combine them into a single dataset, this in turn created a composite key using the original FID and the generated Year column.

Moreover, the group decided to include a camera location dataset in correlation to the yearly camera violations. This decision was chosen to provide a more in-depth analysis on each individual camera. The Camera Locations primary key column, objectID, was found to be the as the yearly datasets FID. This in turn provided additional details on each camera's longitude, latitude, camera direction, installation year, etc.

Thereby, an inner join was performed between the unioned yearly dataset and the camera information dataset to append the related attributes, shown in Fig. 1. The inner join was a necessity as certain cameras were not installed until more recent years.

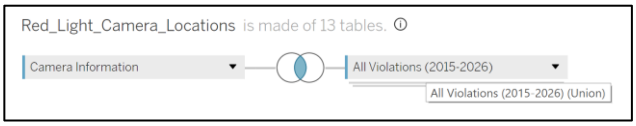

Figure 1. Data model (table relationships) viewed in Tableau.

Finally, the monthly columns for each year were pivoted to create a single Month field name and a Total Violations field attribute. The pivoting process transformed the dataset from 831 rows and 26 columns to 9972 rows and 16 columns, improving its structure for analysis.

VI. DATA EXPLORATION

The data exploration phase evaluated the dataset to determine meaningful analytics for visualization. The intended explorations included calculating the overall violation volume to determine the count, sum, average of the dataset's violations. Identify traffic patterns in the city of Ottawa with the highest and lowest monthly violation intersections. Monthly and yearly trends are analyzed to detect possible seasonal patterns, such as the differences between winter and summer driving behaviour. Additionally, explore the relationship between camera direction and monthly violation frequency to better understand traffic patterns across Ottawa intersections.

Furthermore, throughout the data exploration process, due to the use of pivoting the data, the original Total Violation and Highest Monthly Violations columns were no longer viable. Thereby, the group needs to rely on the newly created pivoted field name and attribute to achieve the expected data visualizations.

VII. DATA VISUALIZATION

A. Overview Dashboard

The Overview dashboard, shown in Fig. 2, serves as the entry point of the interactive Tableau dashboard, providing stakeholders with a consolidated summary of all key performance indicators, metrics, and visualizations required for program-level decision making.

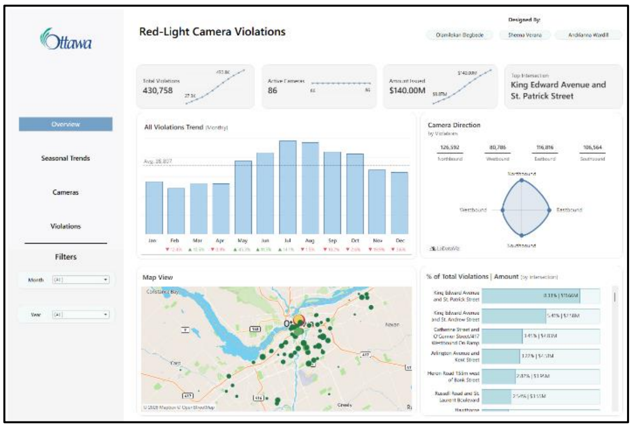

Figure 2. Overview dashboard.

Four KPI cards were designed to display the most critical headline figures. The first displays the total number of violations, computed as the aggregate sum of all red-light infractions recorded across the entire camera network for the selected period. The second card shows the count of active cameras, defined as the distinct number of cameras that recorded at least one violation during the period. It is worth noting that certain camera records contained null or NA values, attributed to temporary deactivation during intersection construction, which were excluded from this count. The third KPI card presents the total amount issued in fines, calculated by multiplying total violations by the standard $325 CAD flat fee per infraction. This figure provides stakeholders with a direct measure of the financial enforcement impact of the program and supports budget justification and public accountability reporting. The fourth card identifies the camera intersection with the highest recorded violations for the selected period, serving as an immediate enforcement priority indicator that updates dynamically as year and month filters are adjusted. Each KPI card is accompanied by a cumulative month-over-month sparkline showing the metric value from the start to the end of the selected period, providing visual trend context alongside the headline figure.

The presented KPI cards can help the city of Ottawa to analyze the strong points in the data. They provide quick overviews before advancing into the charts for a deeper analysis.

A monthly bar chart was included to show how total violations fluctuate across the twelve months of the selected year. A dashed reference line representing the annual monthly average was overlaid on the chart to provide a performance benchmark, making it easier to identify months that exceed or fall below the typical level. Month-over-month percentage change labels were added below each bar to quantify the directional shift relative to the prior month, supporting the identification of seasonal spikes and enforcement gaps.

A radar chart was used to visualize violation volume by camera facing direction, covering the four cardinal orientations: Northbound, Eastbound, Southbound, and Westbound. The polygon arm extending furthest from the center represents the direction capturing the greatest share of city-wide violations. This view enables planners to assess whether specific traffic flow directions exhibit disproportionate red-light compliance issues, informing decisions around camera orientation and roadside signage placement.

A bubble map was included to provide a geographic view of all red-light camera locations across Ottawa. Bubble size encodes violation volume, with larger circles indicating higher-violation intersections. Each bubble is color-coded according to its assigned risk tier, derived from the box plot distribution analysis described in the Violations dashboard section. This view enables stakeholders to identify geographic clustering of high-risk intersections and evaluate whether enforcement coverage is distributed equitably across the city or concentrated along specific corridors.

A horizontal bar chart ranks all intersections by their share of total city-wide violations, with each bar labeled with both the percentage contribution and the corresponding estimated fine revenue. This view highlights which intersections are responsible for the largest proportion of enforcement activity, supporting prioritization decisions around camera maintenance, patrol deployment, and infrastructure review.

Year and month filters were integrated into the dashboard to allow stakeholders to scope all views to a specific time period, enabling targeted analysis without navigating away from the summary page.

B. Seasonal Trends Dashboard

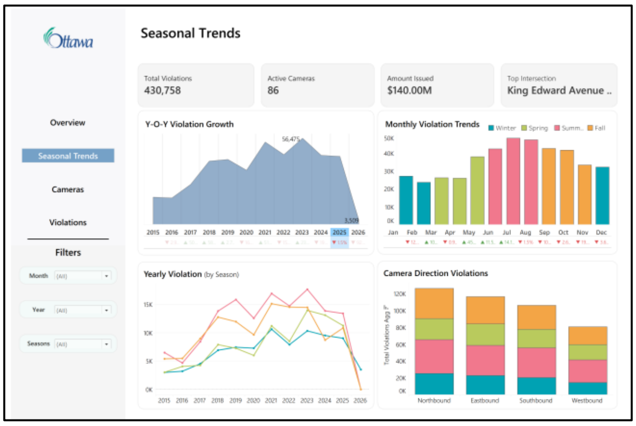

Figure 3. Seasonal trends dashboard.

To implement the seasonal trends dashboard, the four KPIs presented use the same approach as the overview dashboard discussed in Section VIII-A. As seen in Fig. 3 each of the four KPI cards do not contain the cumulative month-over-month sparkline. They represent cumulative changes from the earliest dataset to the most recent dataset. It is more suitable to be included in the overview dashboard, where the objective is to visualize overall dataset progression.

Instead, the season trends dashboard focuses on comparing patterns across seasonal patterns and variations across months and years. This in turn can create analytical confusion, as the sparklines reflect longitudinal growth rather than seasonal fluctuations. These were omitted to avoid creating redundancy and adding unnecessary visual elements that do not contribute additional analytical value.

Although the month-over-month sparklines were not included with the KPI cards, the month-over-month percentage change labels were kept. As they provide a visual indicator of how values changed relative to the period before them. This allows users to quickly assess an increase or decrease in the data without introducing visual elements that may distract the user from the seasonal trend analysis.

The chosen graphs were the monthly and yearly charts that were created during the data visualization stages. The year-over-year line chart was changed to an area line chart for better visibility when reading. This change helps improve visibility by emphasizing the magnitude of changes between years and comparing it to the monthly chart.

Since the goal was to analyze seasonal trends, the months were grouped using a calculated field to create four seasons: winter, spring, summer, and fall. This helps illustrate how violations fluctuate throughout the year, thereby making it easier to identify recurring seasonal changes. This may bring up the discussion of how the harsh winters in Ottawa affect driving behaviours and decrease the amount of red light violations.

Another selected graph from the data visualization stage was the camera direction violations to determine frequency, however, this was later changed to a stacked bar chart based on the seasonal groups. This visualization further supports how violations occur across seasons while also identifying the direction of traffic that contributes to the most amount of violations.

Finally, a new graph was added to display the yearly violations per season. This graph was included during the construction of the dashboard to further examine whether there is a correlation between seasonal weather and red-light violations.

Filters are an important aspect of the dashboard because they assist with analysis. Three filters were selected: year, month, and season. The yearly filter was applied to the monthly violation bar chart and the camera direction violations chart. The month and season filters were applied to the yearly violations per season chart, year-over-year violation growth chart, and the camera direction violations chart. Lastly, a season legend was included to aid viewers with understanding the seasonal groupings. All filters were applied to the first KPI card, this displays the total violations and number of distinct cameras.

C. Camera Insights Dashboard

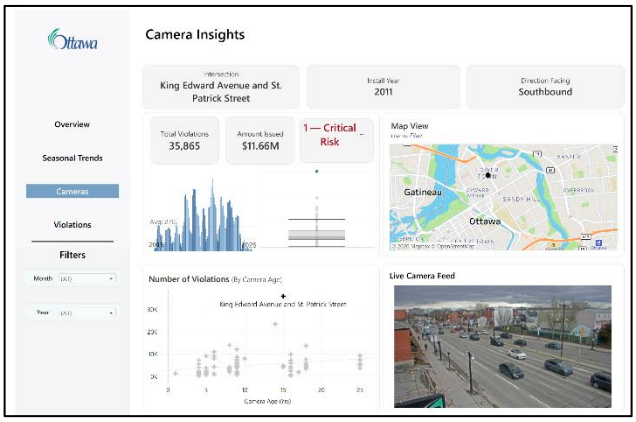

Figure 4. Camera insights dashboard.

The camera insights dashboard shown in Fig. 4 was designed to provide both a high-level summary and a detailed, interactive view of red-light camera activity. KPI cards were used to highlight key information about each camera location. These include the intersection name, install year, and direction the camera is facing, which together specify basic information about the chosen red-light camera.

Additional KPI cards were created to summarize performance metrics. One card displays the total number of violations, while another shows the total amount issued in fines. For the total amount issued, the total amount of violations was multiplied with the $325 fine which is a fine of "$260 plus a $5 service fee and $60 victim surcharge [15]." These values are dynamic. They are responsive to the selected year and month filters, allowing users to analyze trends over specific time periods.

A bar chart was included to show how violations change over time on a monthly basis. An average reference line was added to provide a benchmark, making it easier to compare individual periods against the overall trend. The bar chart displays all months from 2015 to 2026. This allows users to identify patterns and or recurring spikes. Additionally, it aids in comparing the increase of camera installations and how it affects the increase or decrease in red-light violations.

The risk classification indicator was also included on a KPI card to quickly assess the severity of violations at each camera location. This risk box categorizes each intersection into one of five levels, ranging from "1 — Critical Risk" to "5 — Minimal Risk." The classification is based on the average yearly violations for each location. This value is then compared to thresholds from the box plot, such as the median and quartiles, to determine the risk level. The risk classification KPI can help identify locations that may require enforcement review or traffic safety interventions.

The box plot itself is displayed directly on the dashboard. It provides a visual representation of how violations are distributed across cameras. It shows the median, quartiles, and potential outliers. If a camera's average violations exceed the upper whisker, it is labeled as Critical Risk, indicating it is among the highest-performing (or most problematic) locations. Values above Q3 are classified as High Risk, above the median as Moderate Risk, and above Q1 as Low Risk, while anything below Q1 is considered Minimal Risk. This approach standardizes risk assessment across all cameras and allows users to quickly identify intersections that require the most attention.

To explore potential relationships between variables, a scatter plot was created showing the number of violations by camera age. This chart helps assess whether there is any correlation between how long a camera has been installed and how many violations it records. A trend line was added to examine the relationship between camera age and the total number of violations. The regression line is defined as shown in (1).

$$\text{Total Violations} = 201.179 \times \text{Camera Age} + 2798.19 \qquad (1)$$

This shows a slight positive relationship. This means violations tend to increase as camera age increases. However, the low R^2 value (0.0704) indicates the relationship is weak, with camera age explaining only a small portion of the variation. Although the p-value (0.0135) suggests the relationship is statistically significant, other factors likely have a greater impact on violation counts.

A live camera feed component was included to provide real-time visual context. This section updates based on the selected camera from the map. It uses the links found within the Traffic Camera Locations website. There was a limited availability of live camera feeds. The City of Ottawa does not provide live feeds for every intersection included in the dataset, which created gaps in the real-time visualization component. As a result, the live camera feed section could only be populated for a subset of locations, while others remained unavailable.

The map view serves as an interactive feature of the dashboard. Each camera is represented as a selectable point, and clicking on a camera applies a filter across the dashboard. This interaction updates multiple elements at once, including the intersection, install year, direction, total violations, total amount issued, the violations bar chart, the scatter plot, risk indicator, box plot and the live camera feed. This enables users to select a specific location and instantly see all relevant information. The interactive map helps in examining location-based patterns and compares intersections visually.

D. Violations Dashboard

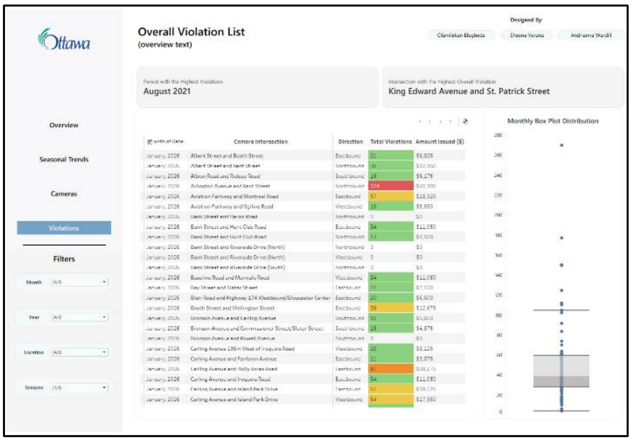

Figure 5. Camera insights dashboard.

The Violations dashboard, shown in Fig. 5, provides stakeholders with a detailed, filterable record of all red-light camera violations across the full 2015–2026 dataset. The view is designed to support both reporting and in-depth investigative analysis, with filters available for month, year, intersection, and season.

Two KPI metric cards anchor the top of the view. The first identifies the month and year combination with the highest recorded violation count across all cameras and all years in the dataset, representing the all-time peak enforcement period. This figure serves as a historical benchmark for program growth presentations and can be used to investigate what conditions, whether seasonal, infrastructural, or related to traffic pattern changes, contributed to the spike. The second card identifies the single camera intersection with the highest cumulative violations across the entire dataset, representing the location of greatest sustained red-light compliance risk in Ottawa and the first location that should be considered in any enforcement prioritization or safety infrastructure review.

The violation records table provides a fully filterable row-level view of all recorded infractions, displaying the month, camera intersection, facing direction, total violations, and estimated fine amount for each entry. The Total Violations column is color-coded according to the risk tier assigned to each intersection: red for Critical Risk, orange for High Risk, yellow for Moderate Risk, and green for Low Risk. This encoding allows users to visually scan the table and identify problem locations without any additional interaction.

A box plot visualization displays the statistical distribution of average monthly violations across all camera intersections. The box spans the interquartile range from Q1 to Q3, representing the middle 50% of intersections. The median is indicated by the line within the box, and the whiskers extend to the boundary of the typical range. Data points beyond the whiskers represent statistical outliers which are the intersections performing significantly above or below the norm for the fleet. This distribution directly informs the risk classification system applied throughout the dashboard: intersections with average violations exceeding the upper whisker are classified as Critical Risk, those above Q3 as High Risk, those above the median as Moderate Risk, those above Q1 as Low Risk, and those below Q1 as Minimal Risk. This standardized approach ensures risk assessment is data-driven and consistently applied across all camera locations, enabling stakeholders to immediately identify intersections requiring urgent attention and those that may be candidates for camera redeployment.

VIII. Discussion and Results

The interactive dashboards provide a comprehensive view of red-light camera activity in Ottawa from 2015 to 2026. The dashboard combines summary metrics with detailed visualizations for further analysis.

The overview dashboard serves as a centralized summary of the dataset. It highlights key indicators. Most notably, there were a total of 430.8 thousand violations throughout the dataset as of January 2026, whereas in January 2015 there were approximately 8 thousand violations. Other KPIs provide insight that there were 55 active cameras in January 2015, and now, as of January 2026, there are 86 active cameras. Additionally, there were approximately 0.27 million dollars CAD issued in red-light violations in January 2015. Now, as of January 2026, there have been 140 million dollars CAD issued in red-light violations.

King Edward Avenue and St. Patrick Street has the most violations, issuing approximately 11.6 million dollars CAD. This value can be confirmed throughout the dashboard based on the top intersection KPI and the map view. The map highlights that the King Edward Avenue and St. Patrick Street intersection has the largest bubble and the most distinct color compared to other intersections. Additionally, violation counts across camera directions are relatively balanced, although northbound traffic has the highest number of violations at approximately 126 thousand.

The seasonal trends dashboard provides additional context by examining how violations change across months, seasons, and years. Based on the visualizations, 2023 recorded the highest violations in terms of year-over-year growth, with approximately 56 thousand violations. In contrast, 2026 has the lowest violations, as it only contains data for the month of January.

The seasonal visualizations, including the yearly violations per season chart, monthly violation trends, and camera direction violations by season, suggest that weather conditions may influence the number of violations. There is a decrease in violations during the winter seasons and an increase during the summer. The camera direction violations further support the analysis that northbound traffic has the highest number of violations overall, as shown in the overview dashboard.

The camera insights dashboard focuses on individual camera performance and detailed location-based analysis. This dashboard helps identify intersections with unusually high violation levels. It is an interactive dashboard that is better suited for analyzing each intersection individually.

The violations dashboard provides a detailed investigative view across the entire dataset. It offers a long-term historical analysis of each data point, including the year and month, intersection, camera direction, violations, and amount issued. The dashboard also provides direct insight that August 2021 had the highest number of violations across all periods.

IX. Conclusion and Future Work

This project shows how combining scattered public datasets into one consolidated dataset can make traffic data much easier to understand and use. By bringing together red-light camera data from 2015 to 2026 and building an interactive Tableau dashboard, users can quickly spot trends, find high-risk intersections, and better understand how violations change over time.

The dashboard offers both a high-level and detailed view for specific intersections. It includes maps, key metrics, seasonal trends, and risk levels to help users explore the data from different angles. The results show that violations vary depending on time, location, and camera direction, with some intersections consistently having higher violation rates. Seasonal patterns also suggest that factors like weather and traffic conditions play a role.

The dashboard turns complex open data into clear, useful insights that support city officials in making more informed traffic safety decisions.

For future work, it would be beneficial to implement true real-time data integration by connecting directly to a dataset or API that provides live violation data, rather than relying on periodically updated datasets from the City of Ottawa. This would enable near real-time monitoring and faster responses to traffic safety issues.

Furthermore, machine learning models, such as time-series forecasting or regression modeling, could be incorporated to predict future violation trends and identify potential high-risk intersections before incidents occur.

Additional data sources could also be integrated, such as the Traffic Collision dataset from the Open Ottawa website, to better understand the underlying factors influencing red-light violations.

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